Marketing Analytics

BADM 534 | Spring 2027 | Section A | CRN 77732

Course Information	Details
Instructor	Nathan Yang
Email	ncyang@illinois.edu
Office Hours	By appointment
Class Meetings	Wednesdays, 15:30-18:20 (3:30-6:20 p.m.)
Location	Wohlers Hall, Room 243
Meeting Dates	January 20-May 5, 2027; no class March 17

Course Description

Marketing decisions require evidence about customer preferences, behavior, and responses to a firm's actions. This course connects those questions to customer measurement, visualization, regression, conjoint analysis, segmentation, customer lifetime value, text and network analysis, experiments, causal inference, prediction, and recommendation systems. Applications inform product design, acquisition, retention, pricing, and marketing budgets.

AI-assisted coding is integrated throughout the course. Students practice specifying a problem, generating and revising Python code, checking calculations, and explaining what the results support. Customer value, data privacy, and regulatory scope enter the discussion alongside commercial outcomes.

Learning Objectives

- Define a managerial decision, the relevant population, and the evidence needed to inform it.
- Construct and assess data, with explicit observation units, denominators, missing values, and reproducible transformations.
- Apply marketing analytics methods and distinguish descriptive patterns, predictive performance, and causal effects.
- Evaluate AI-assisted code through independent calculations, appropriate validation, and substantive interpretation.
- Communicate recommendations with clear figures, uncertainty, customer consequences, and supporting evidence.

Class Format and Materials

Meetings combine instructor-led explanations, worked examples, Python exercises, and discussion. The course has 15 meetings of 170 minutes. The topic sequence includes two project discussion and presentation sessions; presentation requirements will be specified separately.

The course repository provides lecture notes, session slides, and data with runnable Python. Markdown companions make the materials searchable and usable with LLMs. Resource-library recommendations provide background; assigned readings will be identified separately.

Cases include mHealth food logging, restaurant availability, generative-AI demand, and earnings-call pricing communication. Synthetic examples supply hand-checkable calculations.

Meeting Schedule

All meetings are on Wednesdays, 15:30-18:20, in Wohlers Hall, Room 243.

Meeting	Date	Topics
1	January 20	M1: Marketing decisions and AI-assisted Python
2	January 27	M2: Customer data, measurement, missingness, and earnings-call coverage
3	February 3	M2: Visualization, data quality, and nutrition-informed assessment
4	February 10	M3: Regression and marketing interpretation
5	February 17	M4: Conjoint analysis and product design
6	February 24	M4: Segmentation and customer lifetime value
7	March 3	M5: Text, social media, networks, and pricing communication
8	March 10	M1-M5: Project evidence and midterm discussion/presentations
No Class	March 17	Spring break
9	March 24	M6: Experiments and incremental value
10	March 31	M6: Causal comparisons, policy evaluation, and event windows
11	April 7	M7: Prediction, validation, and marketing actions
12	April 14	M7: Recommendation systems and customer goals
13	April 21	M8: Adaptive recommendations and marketing budgets
14	April 28	M8: Generative AI, consumer choice, pricing, and regulation
15	May 5	M1-M8: Integrated decisions and final discussion/presentations

Holidays and Breaks

Spring break runs from March 13 at 1:00 p.m. through March 22 at 7:00 a.m.; there is no class on March 17. Martin Luther King Jr. Day falls on Monday, January 18, before the first meeting. Regular instruction ends May 5, followed by reading day on May 6. No regular course meetings are scheduled during the May 7-13 final examination period. Any final examination arrangements will be specified separately.

The dates follow the Illinois Senate 2026-2027 academic calendar and the Spring 2027 course listing.

Assessments and Course Policies

Assessment components, grading weights, submission deadlines, presentation requirements, attendance and late-work rules, and assessment-specific AI permissions are to be announced. Contact Nathan Yang at ncyang@illinois.edu for questions or to arrange office hours.